

# Companion to Bird & de Moor, Algebra of Programming

## §10. Greedy Algorithms

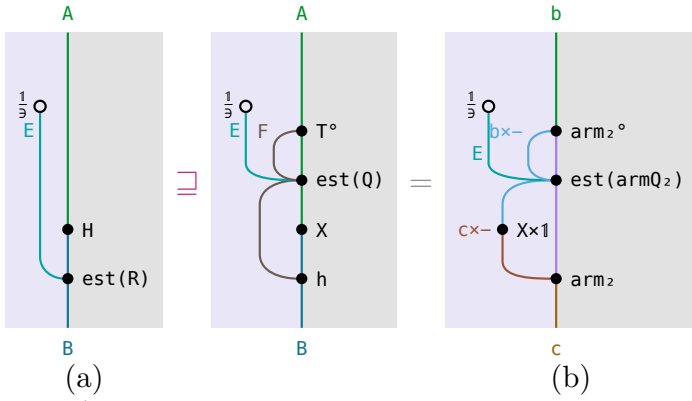
### §10.1. Theory

(10.1a)

$h, T, R, H, M$  as in  $\triangle H$ ; additionally  $Q$  a **connected** preorder on the sets  $\frac{T^\circ}{\exists}$  returns, so that  $\frac{T^\circ}{\exists} \text{est}(Q)$  is entire.

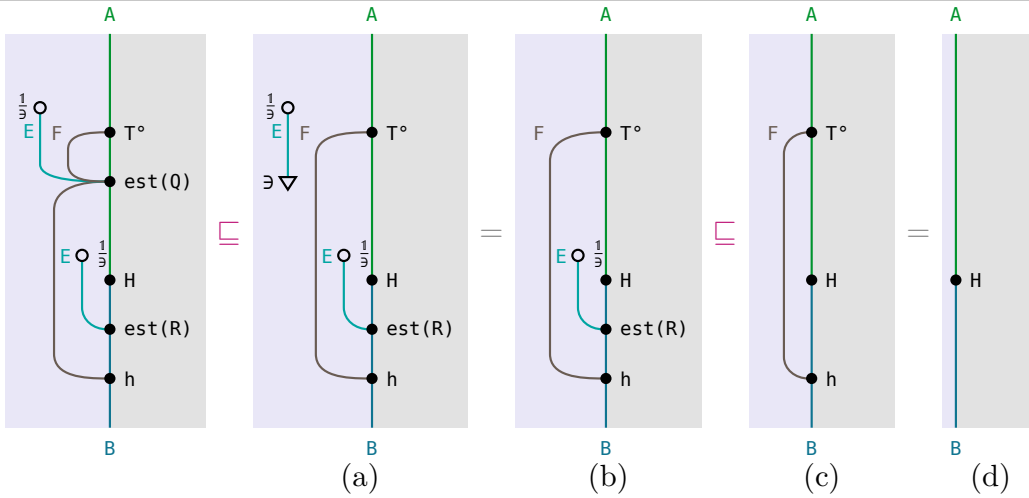
**Map(h)  $\wedge$  MonoAlg(h,R)  $\wedge$  RR $\subseteq$ R  $\wedge$  ThinCondition(T,h,R,Q)  $\Rightarrow$  ( $\mu X : \frac{T^\circ}{\exists} \text{est}(Q)F(X)h$ ) $\subseteq \frac{H}{\exists}$  **est(R)**** Theorem 10.1

the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried and the recursion runs on values alone



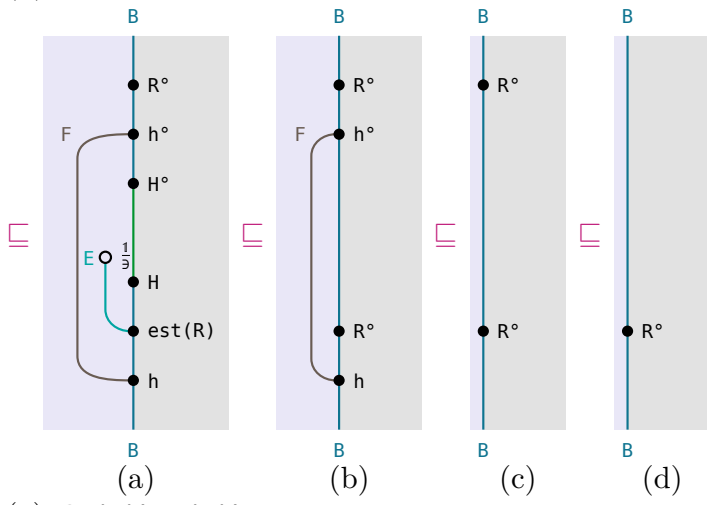
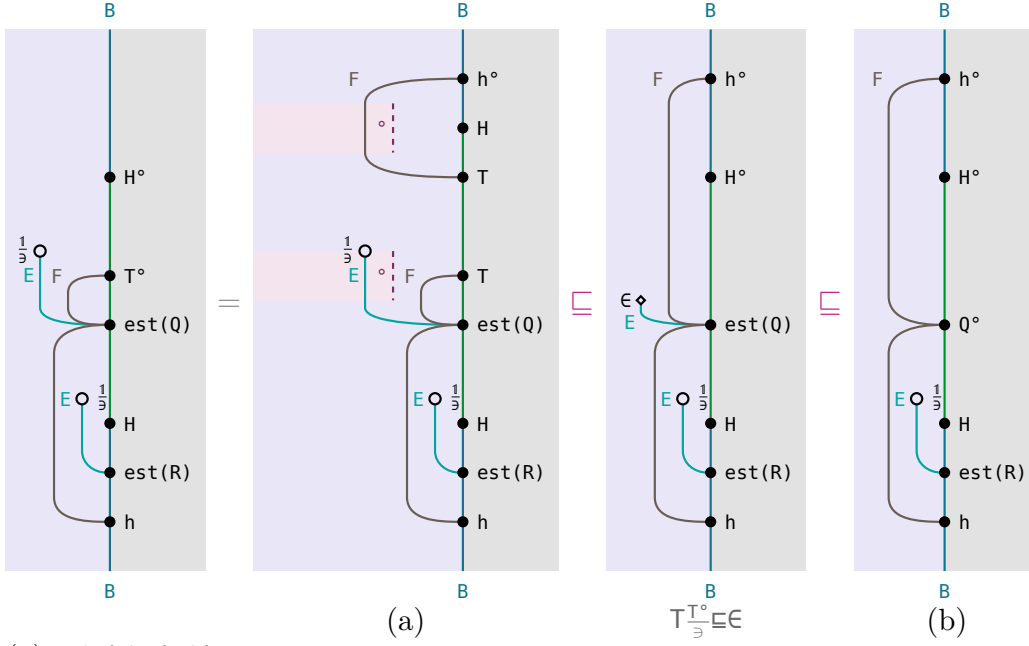
**$T^\circ F(H)h=H \Rightarrow \frac{T^\circ}{\exists} \text{est}(Q)F(\frac{H}{\exists} \text{est(R)})h \subseteq H$**  (10.1b)

taking the input apart in one  $Q$ -extreme way, solving the parts by  $M$  and assembling by  $h$  returns only what  $H$  returns



$$\text{Map}(h) \wedge \text{MonoAlg}(h, R^\circ) \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge T^\circ F(H)h = H \wedge QF(H)h \sqsubseteq F(H)hR \Rightarrow H^\circ \frac{T^\circ}{\exists} \text{est}(Q)F(\frac{H}{\exists} \text{est}(R))h \sqsubseteq R^\circ \quad (10.1c)$$

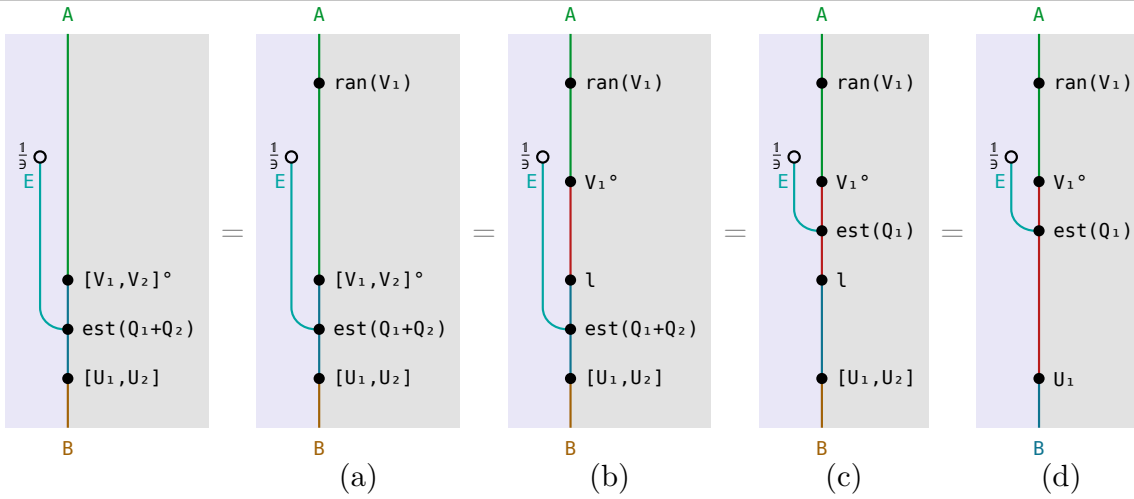
$H^\circ$  followed by the greedy body at  $M$  is  $\sqsubseteq R^\circ$ : an answer of the body is never worse than an answer of  $H$  to the same input



$$V_2 \quad V_1^\circ = \emptyset \Rightarrow \frac{[V_1, V_2]^\circ}{\exists} \text{est}(Q_1 + Q_2)[U_1, U_2] = \text{ran}(V_1) \frac{V_1^\circ}{\exists} \text{est}(Q_1)U_1 \cup \text{ran}(V_2) \frac{V_2^\circ}{\exists} \text{est}(Q_2)U_2$$

Proposition 10.1

when  $V_1$  and  $V_2$  have disjoint ranges, the greedy step over  $[V_1, V_2]$  runs the  $V_1$  step on inputs  $V_1$  reaches and the  $V_2$  step on inputs  $V_2$  reaches



## §10.2. The detab-entab problem

(10.2a)

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$  over  $\text{snoc}$ -lists,  $\alpha \triangleq [\text{nil}, \text{snoc}]$ ,  $H = \text{detab}^\circ$ ;  $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$ ,  $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$ .

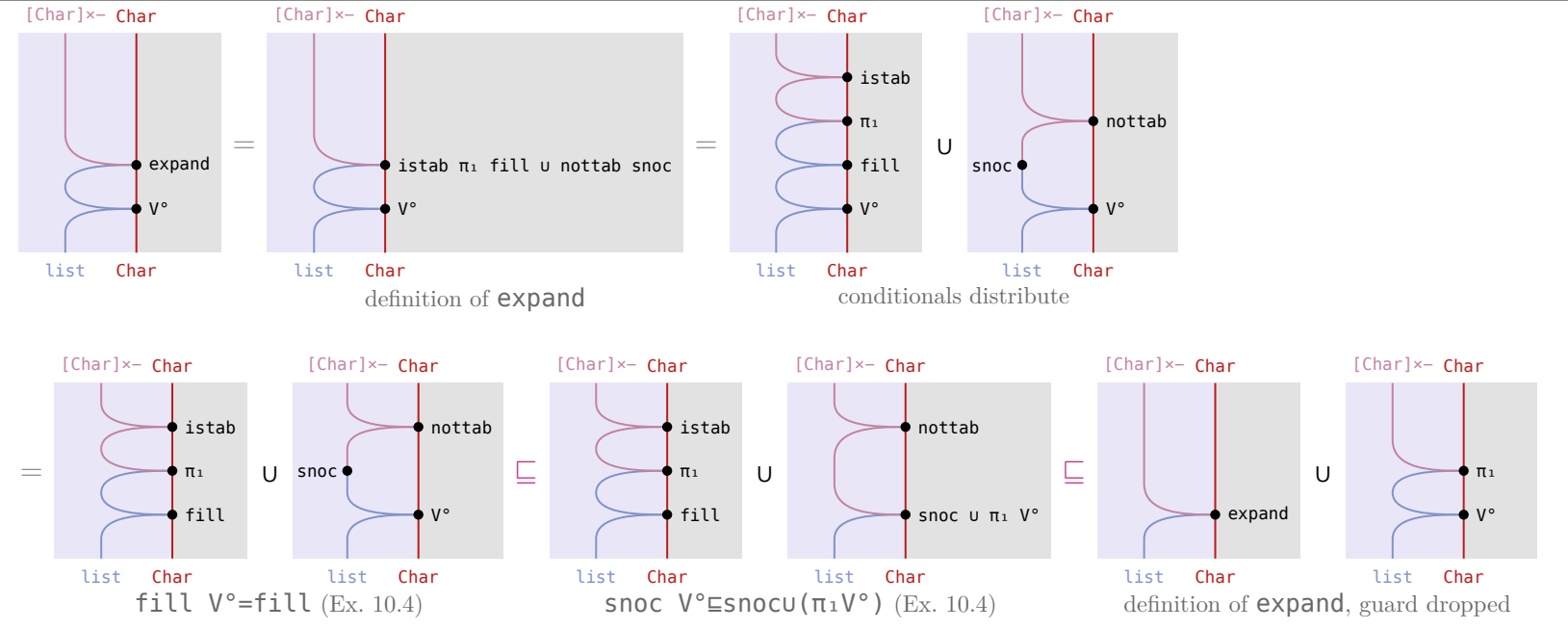
$\text{col} \triangleq ([\text{zero}, \text{count}])$ ,  $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c + 1)$ ; TB the tab, BL the blank, NL the newline, tab stops every  $n$  columns.

$R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U$  the preorder with  $a \sqcup b \iff a = \text{TB} \vee a = b$ ,  $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$ ,  $Q \triangleq 1 + (V \times U)$ .

$\text{unfill } xs$  the shortest prefix of  $xs$  with  $\text{fill}$  ( $\text{unfill } xs = \text{fill } xs$ );  $\text{tbc}$  the trailing blank count,  $\text{triple} \triangleq (\text{unfill } \text{entab}, (\text{tbc}, \text{col}))$ .

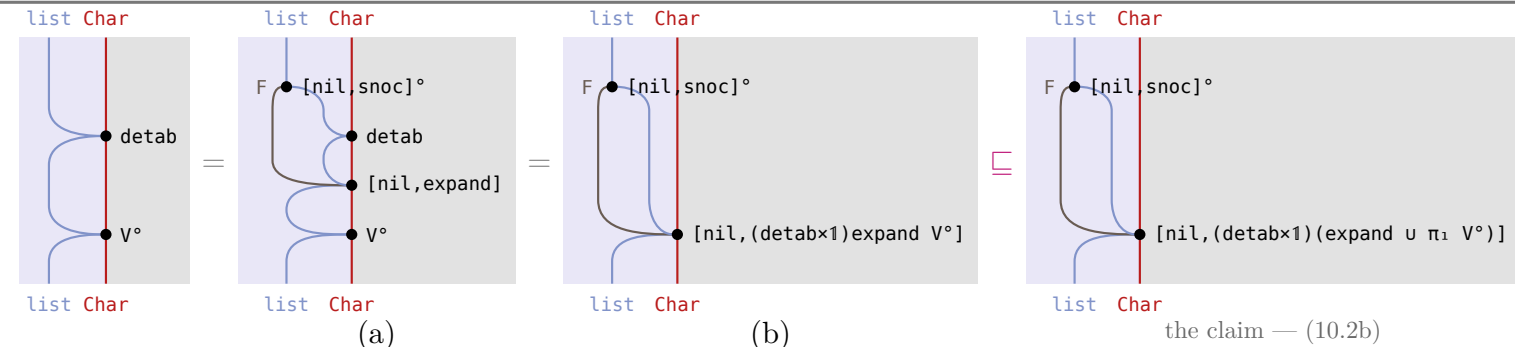
$$0 < n \wedge \text{blank} \neq \text{nl} \implies \text{expand } V^\circ \sqsubseteq \text{expand} \sqcup \pi_1 V^\circ \quad (10.2b)$$

shortening the output of one **expand** step to a  $V$ -smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string ( $\pi_1 V^\circ$ )



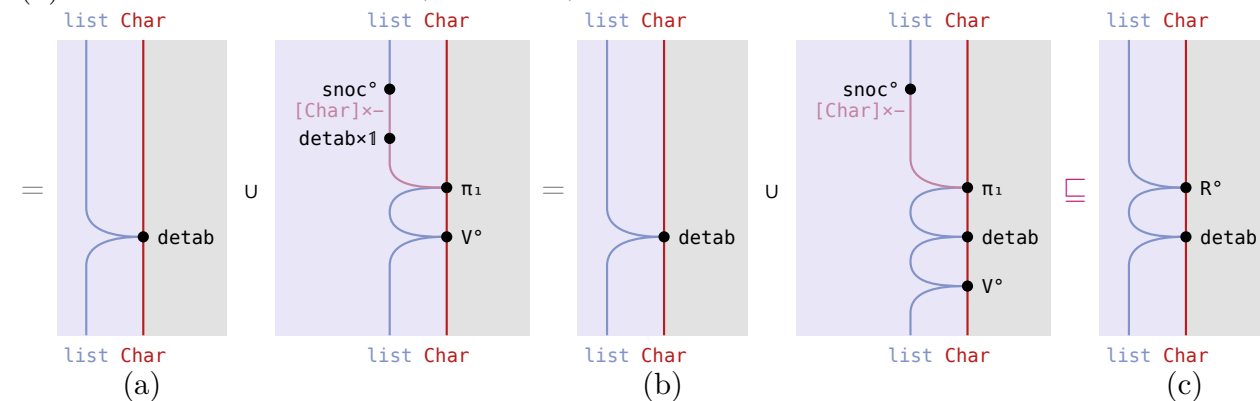
$$0 < n \wedge \text{blank} \neq \text{nl} \implies \text{detab } V^\circ \sqsubseteq R^\circ \text{detab} \quad (10.2c)$$

any  $V$ -shortening of **detab**'s output is **detab**'s output on an input no longer than the given one



(a)  $\text{detab}$  is a fold:  $\text{detab} = \alpha^\circ F (\text{detab}) [\text{nil}, \text{expand}] \triangleq \text{detab}$

(b) coproducts, and  $\text{nil } V^\circ = \text{nil}$  (Exercise 10.4)



(a) distributing  $U$ ; the fold again, and the definition of  $F$

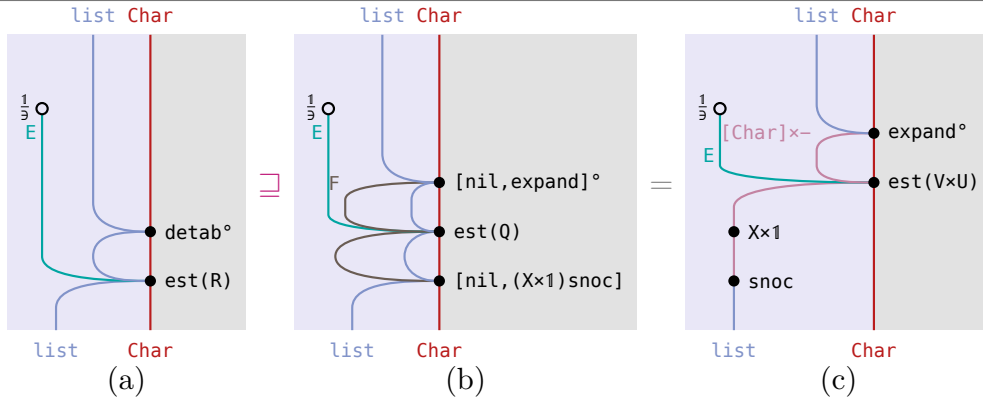
(b) naturality of  $\pi_1$ :  $(\text{detab} \times 1) \pi_1 = \pi_1 \text{detab}$ ;  $\text{snoc}^\circ \pi_1$  is  $\text{init}$

(c)  $\text{init}$  is inductive, so  $X \triangleq \text{detab } V^\circ$ , a solution of  $X \sqsubseteq \text{detab} \cup (\text{init } X)$ , lies below the greatest one,  $\text{prefix detab}$  (induction on the input);  $\text{prefix} \sqsubseteq R^\circ$

(10.2a)

$$0 < n \wedge \text{blank} \neq n1 \Rightarrow (\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{est}(Q)F(X)[\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{3} \text{est}(R) \quad (10.2d)$$

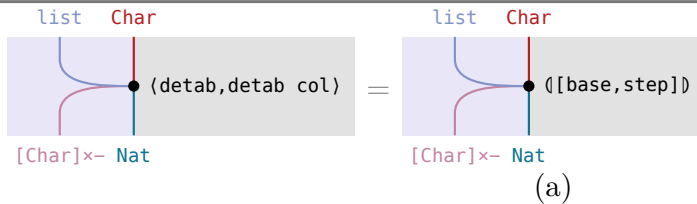
the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and cashing the held blanks in for a tab wherever the column reaches a tab stop



- (a) the specification —  $\triangleq \text{detab}; \text{detab} \text{ entab} = 1$  and nothing shorter does  
 (b) at  $Q \triangleq 1 + (V \times U)$ : one character of input is decided at each step.  $F(T, R) \alpha \alpha R$ .  $\text{detab prefix} \sqsubseteq R^\circ$   $\text{detab}$  is FALSE, — at  $n=8$ ,  $\text{detab} [a, b, c, d, e, TB] = [a, b, c, d, e, BL, BL, BL]$ , whose prefix  $[a, b, c, d, e, BL, BL]$  is longer than any input giving it, and  $\text{detab } V^\circ \sqsubseteq R^\circ$   $\text{detab}$  holds.  $\text{expand } V^\circ \sqsubseteq \text{expand } U$  ( $\pi_1 V^\circ$ ) — shortening the output either leaves the last step alone or discards it  
 (c) Proposition 10.1:  $\text{nil}$  and  $\text{expand}$  have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop

$$\text{blank} \neq n1 \wedge \text{tb} \neq n1 \Rightarrow \langle \text{detab}, \text{detab col} \rangle = \langle [base, step] \rangle \quad (10.2e)$$

one pass over the input carries the output so far together with its column



- (a) both components are folds over the same input;  $\text{step}$  on  $(\text{detab } x, \text{col}(\text{detab } x))$  is  $(\text{detab}, \text{col detab})$  one character further — not a tabulated row

Exercise 10.1

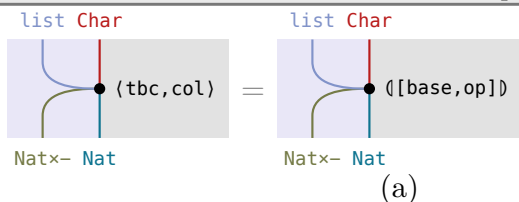
- $\text{blank} \neq n1 \wedge \text{tb} \neq n1 \Rightarrow \langle [base, step] \rangle (\text{convert}(l), r) \Leftrightarrow r = \text{loop}(\text{step}, (([ ], 0), l))$   
the fold of  $[base, step]$  over the input converted to a snoc-list is the left loop of  $\text{step}$  from  $base$
- $\pi_1(\text{loop}(\text{step}, (([ ], c_0), x))) = \text{loop}'((f, g), (c_0, x))$   
Exercise 10.1: when  $\text{step}$  only appends  $f(c, a)$  to the output and moves the state by  $g$ , the output of the loop is  $\text{loop}'(f, g)$ , which never carries the output

- $a \neq \text{tb} \Rightarrow \text{expand}(y, b) = \text{snoc}(x, a) \Leftrightarrow \text{fill}(y) = \text{snoc}(x, a) \wedge b = \text{tb} \vee (y, b) = (x, a)$   
 $\text{expand}$  produces  $x \# [a]$  from a tab after a string that fills to it, or from  $x$  and  $a$
- $0 < n \wedge \text{blank} \neq n1 \Rightarrow (\exists y. \text{fill}(y) = \text{snoc}(x, a)) \Leftrightarrow a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n = 0$   
a string is a  $\text{fill}$  exactly when it ends in a blank on a tab stop

- $\text{entab}(x) = \text{entab}(\text{unfill } x) ++ \text{blanks}(\text{tbc } x)$   
(10.1): the output of  $\text{entab}$  is that of  $\text{unfill } x$  followed by the trailing blanks held back
- $\text{tbc}[] = 0$   
the empty string has no trailing blanks
- $\text{tbc}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n \neq 0 \text{ then } \text{tbc } x + 1 \text{ else } 0$   
a blank off a tab stop adds one held blank; anything else releases them all

$$\text{blank} \neq n1 \Rightarrow \langle \text{tbc}, \text{col} \rangle = \langle [base, op] \rangle \quad (10.2g)$$

one pass counts the held blanks and the column together

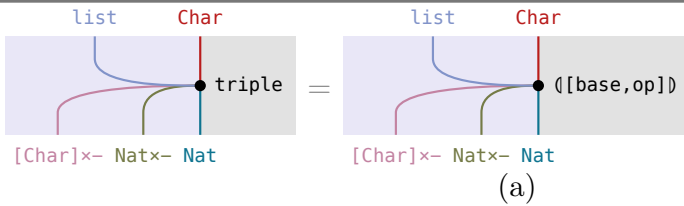


- (a)  $op$  on  $(\text{tbc } x, \text{col } x)$  is  $(\text{tbc}, \text{col})$  one character further, from the equations for  $\text{tbc}$  — (10.1)

$$\text{blank} \neq \text{nl} \Rightarrow \text{triple} = ([\text{base}, \text{op}])$$

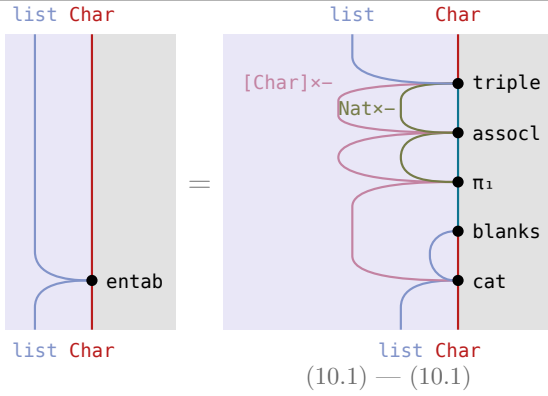
(10.2h)

one pass carries the output up to the held blanks, their count and the column

(a)  $\text{op}$  on  $\text{triple } x$  is  $\text{triple}$  one character further: (10.1) where the blanks are released — (10.1)

$$\text{entab} = \text{triple} \text{ assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$$

(10.2i)

 $\text{entab}$  is  $\text{triple}$  with the held blanks appended to its output

### §10.3. The minimum tardiness problem

(10.3a)

$\text{FX} = 1 + (\text{X} \times \text{Job})$ ,  $\alpha \triangleq [\text{nil}, \text{snoc}]$  on schedules,  $\beta \triangleq [\text{nil}, \text{snag}]$  on bags,  $\text{snag}$  putting a job into a bag;  $\text{bagify} \triangleq ([\beta]) : [\text{Job}] \rightarrow \text{Bag Job}$ ,  $\text{H} = \text{bagify}^\circ$ .

$\text{ct}, \text{dt}, \text{wt} : \text{Job} \rightarrow \text{Real}$  the completion, due and weighting quantities of a job;  $\text{penalty}(\text{xs}, j) = (\text{sum}(\text{list}(\text{ct}) (\text{xs})) + \text{ct}(j) - \text{dt}(j)) \times \text{wt}(j)$ .

$\text{cost} \triangleq \frac{\text{prefix}}{\sup} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{ est}(\geq)$ ,  $\text{cost } [] = 0$ ,  $\text{cost } (\text{xs} \# [j]) = \text{bmax } (\text{cost } \text{xs}, \text{penalty } (\text{xs}, j))$ ,  $\text{R} \triangleq \text{cost} \leq \text{cost}^\circ$ .

$\text{perm} \triangleq \text{bagify } \text{bagify}^\circ = ([\text{nil}, \text{add}])$ ,  $\text{add } (\text{xs}, j) = \text{ys} \# [j] \# \text{zs}$  for some  $\text{xs} = \text{ys} \# \text{zs}$ .

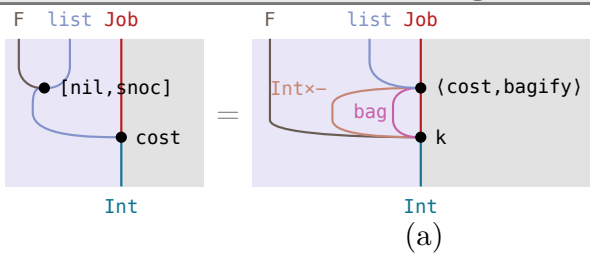
$\text{k} \triangleq [\text{zero}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}]$ ,  $\text{f} \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$ ,  $\text{Q} \triangleq \text{f} \leq \text{f}^\circ$ ,  $\text{Q}' \triangleq (\text{bagify}^\circ \times 1) \text{ penalty} \leq \text{penalty}^\circ (\text{bagify} \times 1)$ .

$\text{g} \triangleq [\text{zero}, \text{penalty}]$  (10.5),  $\text{m} \triangleq [\text{zero}, \pi_1 \text{ cost}]$  (B&dM's  $h$ , renamed as in (10.3e)) (10.6).

$$[\text{nil}, \text{snoc}] \text{ cost} = \text{F}((\text{cost}, \text{bagify})) \text{ k}$$

(10.3b)

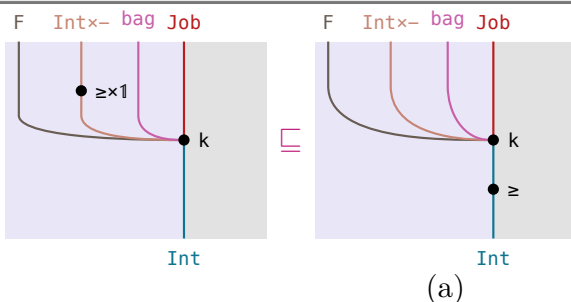
the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job

(a)  $\text{cost } (\text{xs} \# [j]) = \text{bmax } (\text{cost } \text{xs}, \text{penalty } (\text{xs}, j))$ , and  $\text{penalty}(\text{xs}, j)$  reads only the bag of  $\text{xs}$  —  $\triangleq \text{bagify}$ 

$$\text{F}(\geq \times 1) \text{ k} \leq \text{k} \geq$$

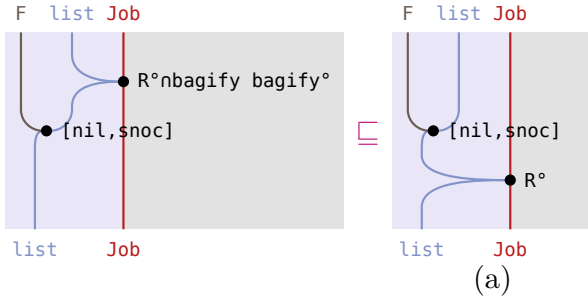
Exercise 10.5

a larger cost before the last job gives a larger cost after it

(a)  $\text{bmax}$  is monotone in its first argument

$$F(R^\circ \text{nbagify bagify}^\circ)[\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}]R^\circ \quad (10.2)$$

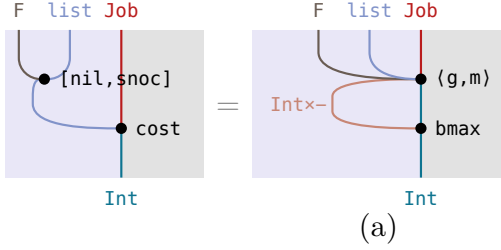
improving the schedule before the last job, among schedules of the same bag, improves the whole schedule



(a) Proposition 9.3 at  $S \triangleq \text{bagify}$ ,  $\leq \triangleq \text{dp-context-mono}$ , with (10.3b) and Exercise 10.5

$$[\text{nil}, \text{snoc}] \text{cost} = (g, m) \text{bmax} \quad (10.4)$$

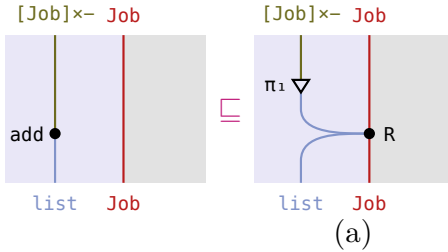
the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it



(a)  $\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$  and  $\text{bmax}$  commutes —  $\triangleq \text{bagify}$

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow \text{add} \sqsubseteq \pi_1 R \quad (10.7)$$

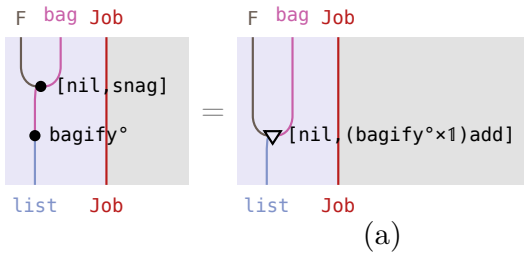
putting one more job anywhere into a schedule never lowers its cost



(a) induction on where  $j$  goes: every job after it starts later, so no penalty falls

$$[\text{nil}, \text{snag}] \text{bagify}^\circ = [\text{nil}, (\text{bagify}^\circ \times 1) \text{add}] \quad (10.8)$$

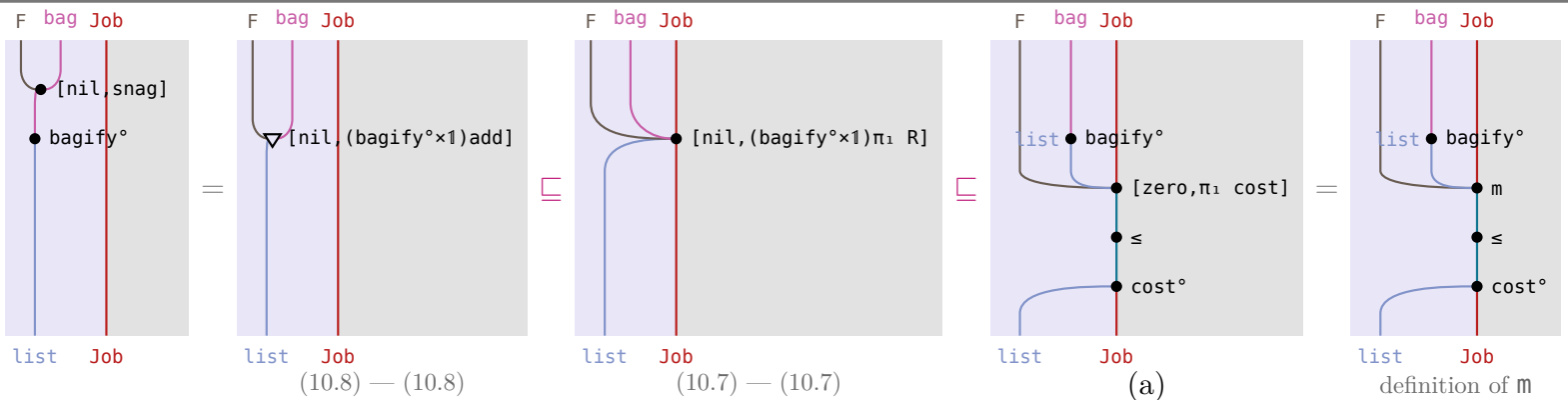
the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere



(a) take out the last copy of  $j$  and put it back —  $\triangleq \text{bagify}$

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow [\text{nil}, \text{snag}] \text{bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) m \leq \text{cost}^\circ \quad (10.3c)$$

an ordering of a bag with one more job costs at least the cost of the ordering of the bag before its last job

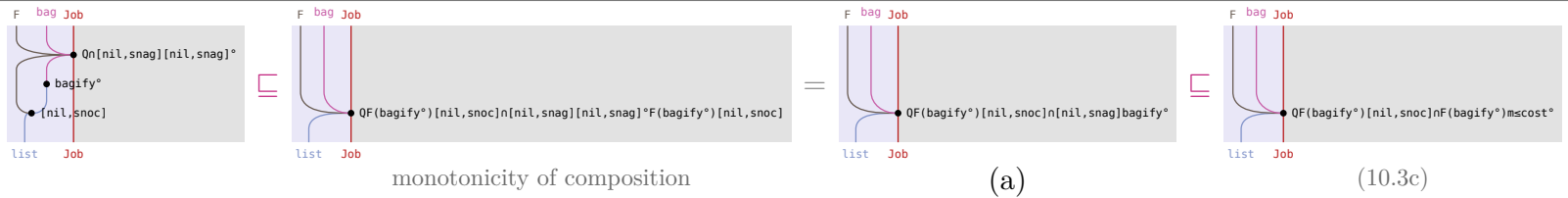


(a) definition of  $R$ , and  $\text{nil} \sqsubseteq \text{zero} \leq \text{cost}^\circ$

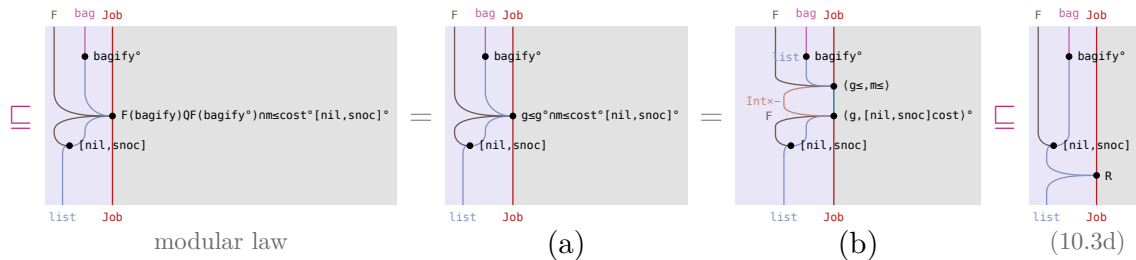
$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (Qn[\text{nil}, \text{snag}][\text{nil}, \text{snag}]^\circ F(\text{bagify}^\circ)[\text{nil}, \text{snoc}] \sqsubseteq F(\text{bagify}^\circ) [\text{nil}, \text{snoc}]R) \quad (10.3)$$

$[\text{nil}, \text{snoc}]R$

a schedule of a bag ending in a job of least penalty is no worse than any schedule of the same bag



(a)  $\beta^\circ F(\text{bagify}^\circ)\alpha = \text{bagify}^\circ$ , since  $\text{bagify} = (\beta)$

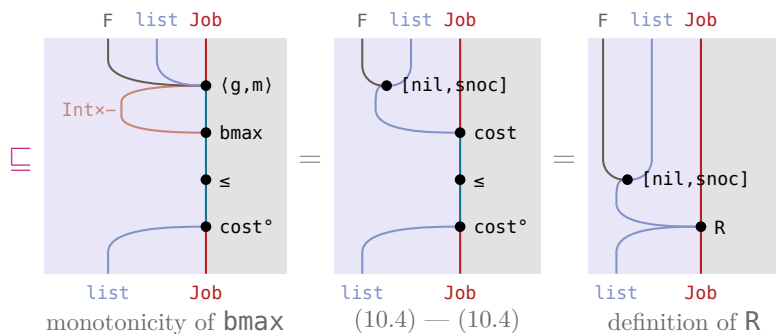
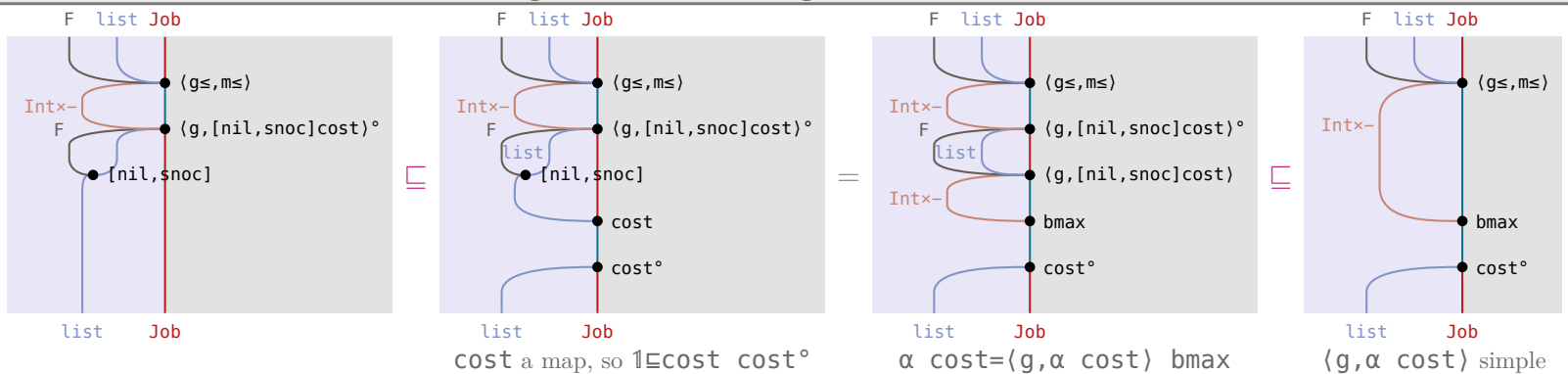


(a) choice of  $Q$ :  $F(\text{bagify}) \ Q \ F(\text{bagify}^\circ) = g \leq g^\circ$

(b) products:  $\langle R, S \rangle \langle T, U \rangle^\circ = RT^\circ \cap SU^\circ$

$$\langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{cost} \rangle^\circ [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}]R \quad (10.3d)$$

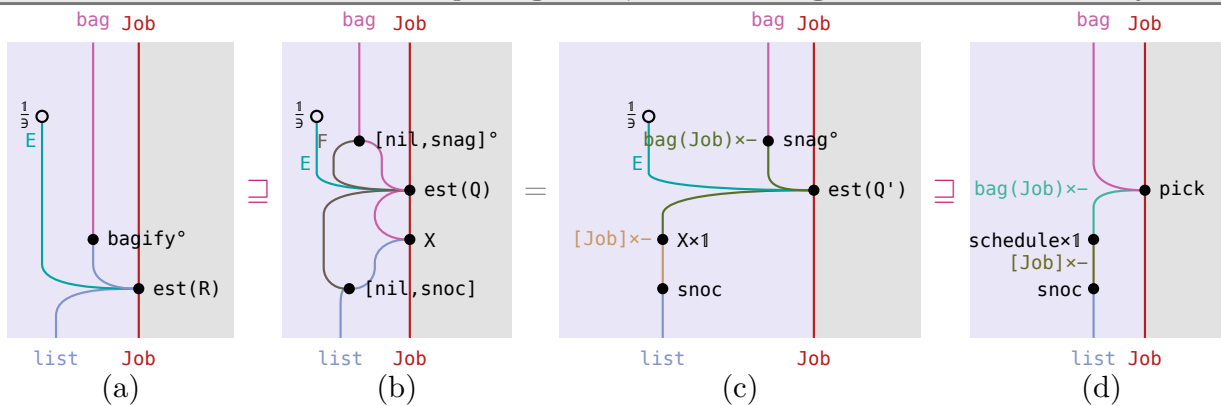
a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job, gives a schedule costing at most the whole



$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \frac{[\text{nil}, \text{snag}]^\circ}{3} \text{est}(Q)F(X)[\text{nil}, \text{snoc}] \sqsubseteq \frac{\text{bagify}^\circ}{3} \text{est}(R), \quad (10.3e)$$

**schedule = (null-nil, pick (schedule×1) snoc)**

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag, putting it last, and scheduling what is left the same way



(a) the specification —  $\triangleq \text{bagify}$

(b) No greedy **reduce** exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole

(c) Proposition 10.1:  $\text{nil}$  and  $\text{snag}$  have disjoint ranges

(d)  $\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{3} \text{est}(Q')$ , a partial function, quadratic in the number of jobs

## §10.4. The TeX problem

(10.4a)

$\text{interval} \triangleq \text{val} \text{ round} : \text{Decimal} \rightarrow [0, 2^{16})$ ,  $\text{val} \triangleq ([\text{zero}, \text{shift}])$ ,  $\text{shift } (d, r) = (d+r)/10$ ,  $\text{round } r$  rounds  $2^{16}r$  to the nearest integer:  $\text{round } r = n \iff 2n-1 < 2^{17}r < 2n+1$ .

$\text{interval } n = ((2n-1)/2^{17}, (2n+1)/2^{17})$ ,  $r \text{ inrange } (a, b) \iff a < r < b$ ,  $\text{round}^\circ = \text{interval inrange}$ ,  $R \triangleq \text{length} \leq \text{length}^\circ$ .

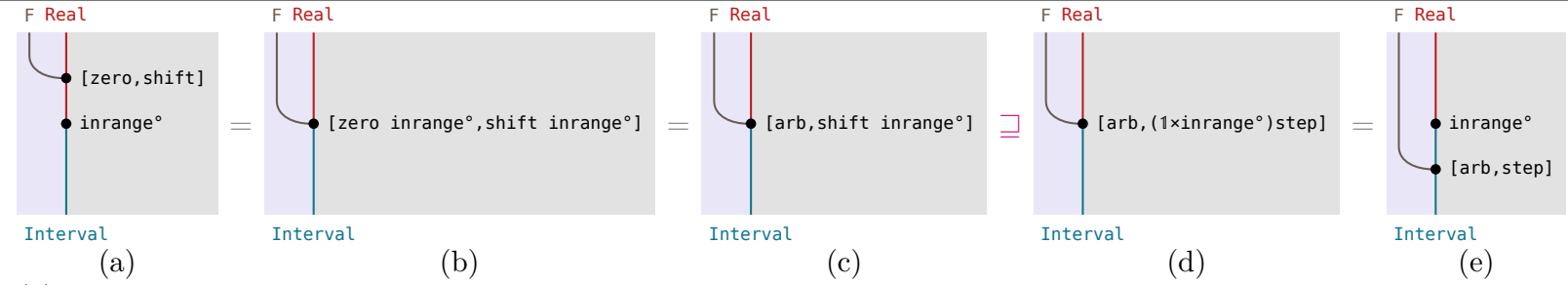
Interval the pairs  $(a, b)$  with  $\text{Legal}(a, b) \iff 0 < b < 1$  and  $a < b$  (10.9);  $\text{step\_legal}$  is what types  $\text{step } [\text{arb}, \text{step}] : 1 + (\text{Digit} \times \text{Interval}) \rightarrow \text{Interval}$ ,  $\text{step } (d, (a, b)) = ((d+a)/10, (d+b)/10)$ .

$\text{FX} = 1 + (\text{Digit} \times X)$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ ,  $H \triangleq ([\text{arb}, \text{step}])^\circ$ ,  $! : \text{Digit} \times \text{Interval} \rightarrow 1$ ,  $Q \triangleq (l^\circ !^\circ r) \cup 1$   $l, r$  are coprod-laws's injections into  $\text{FX} = 1 + (\text{Digit} \times X)$ , so  $l : 1 \rightarrow \text{FX}$  and  $r : \text{Digit} \times X \rightarrow \text{FX}$ ,  $w \triangleq 2^{17}$ .

 $([\text{arb}, \text{step}])^\circ \text{Eval inrange}^\circ$ 

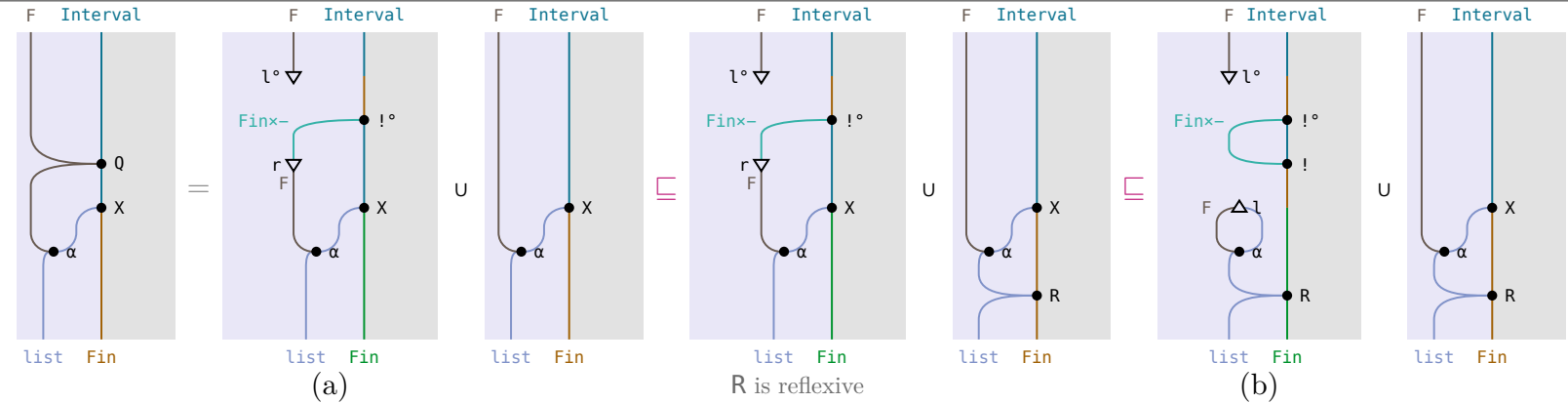
(10.4b)

every interval got by folding a decimal's digits with  $[\text{arb}, \text{step}]$  has the decimal's value strictly inside it

 $QF(X) \alpha EF(X) \alpha R$ 

(10.4c)

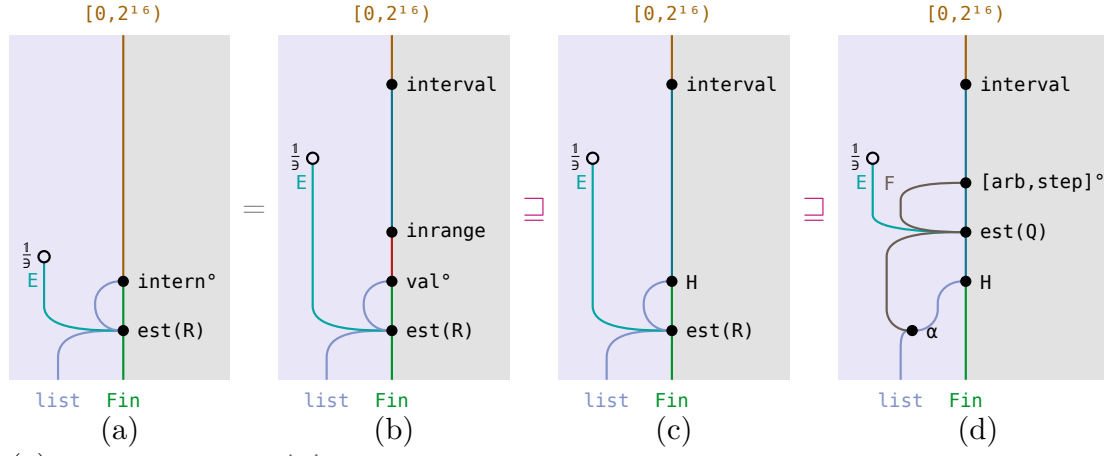
choosing by  $Q$  among the one-step decompositions before building with  $F(X)$  and  $\alpha$  yields only decimals that are  $R$ -related to one built without choosing



(a)  $!^\circ !^\circ \sqsubseteq 1$  on  $1$ : universal property of  $!$   
 (b)  $l F(X) = l$ : definition of  $F$   
 (c)  $l$  is simple:  $l^\circ l \sqsubseteq 1$   
 (d)  $U$  is idempotent

$$\text{interval}(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q)F(X)\alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{est}(R) , \quad \text{extern}(n)=f(2n-1, 2n+1) \quad (10.4d)$$

a shortest decimal whose internal representation is the given multiple of  $2^{-16}$  is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do



- (a) the specification —  $\triangleq$  **intern**  
(b) **round**<sup>°</sup> is not a map, but **interval** is, so it comes out of the transpose  
(c) the type restriction (10.9): a shortest decimal **H** gives an interval is a shortest one among all decimals inside it, and is inside it by (10.4b)  
(d)  $\frac{[\text{arb}, \text{step}]^\circ}{3}$  returns at most two elements — stop, or take one more digit — and ! **nil**<sub>cons</sub> **R**<sup>°</sup> makes it stop whenever stopping is legal

- $f(a,b)=x \iff a < 0 \wedge x=[] \vee \neg a < 0 \wedge (\exists d. \exists y. d = \lfloor 10b \rfloor \wedge f(10a-d, 10b-d)=y \wedge x=[d]\#y)$   
 $f$  gives  $(a,b)$  the empty decimal when  $a < 0$ , and otherwise the digit  $d = \lfloor 10b \rfloor$  followed by what it gives  $(10a-d, 10b-d)$  (10.4e)
- $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q)F(X)\alpha) = f$   
the least solution of the greedy recursion of (10.4d) is  $f$
- $d = \lfloor 10b \rfloor \wedge e = \lfloor 10b \rfloor \implies d=e$   
Exercise 10.12: two digits  $d$  and  $e$  with  $0 < 10b-d < 1$  and  $0 < 10b-e < 1$  are equal
- $(0 < q-p \wedge q < w) \wedge p < 0 \implies f(p,q)=[]$   
the program's first clause:  $f(p,q)=[]$  when  $p < 0$
- $(0 < q-p \wedge q < w) \wedge \neg p < 0 \wedge d = \lfloor 10q/w \rfloor \implies f(p,q)=[d]\#f(10p-wd, 10q-wd)$   
the program's second clause: with  $d = (\lfloor 10q \rfloor \text{ div } w)$ ,  $f(p,q)$  is  $d$  followed by  $f(10p-w \cdot d, 10q-w \cdot d)$
- $(0 < q-p \wedge q < w) \wedge \text{Reach}(p,q) \implies f(p/w, q/w) = f(p,q)$   
on the pairs the program reaches from **interval**  $n$ , the integer  $f$  at  $(p,q)$  is a value of the rational  $f$  at  $(p/w, q/w)$
- **interval**  $f = \text{extern}$   
 $f$  after **interval** is the program **extern**, whose value at  $n$  is  $f(2n-1, 2n+1)$

## §11. Appendix

### §11.1. $P(S) \text{ est}(R) = (\exists S) n(\epsilon \setminus (SR^\circ))$

(11.1a)

$(p, q)$  tabulates  $W \triangleq (\exists S) n(\epsilon \setminus (SR^\circ))$ , and  $y \triangleq \frac{(p \exists S) n(qR)}{\exists}$ , a map.

(11.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq yy^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(11.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f^\circ, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) n(qR) \end{array}} \xleftarrow{\Delta \neg n, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

(11.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, \circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple} \\ \\ \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg n, \cdot T \neg / T, \circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array} \\ \\ \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple} \\ \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) n(qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq qq^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half} \end{array}$$

### §11.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$

(11.2a)

$$\begin{array}{c} \boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)} \\ \\ \xrightarrow{P \text{ a relator}} \boxed{P(\text{dom}(\text{est}(R))) P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{est-711}} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \end{array}$$

(11.2b)

$$\begin{array}{c} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ P(\text{est}(R)) \text{ est}(R) \end{array}} \xleftarrow{(11.1b) \text{ at } S := \text{est}(R), \Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \\ \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \xleftarrow{P(\text{dom}(\text{est}(R))) \sqsubseteq 1} \boxed{\begin{array}{c} \text{union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \\ \xleftarrow{T(UnV) \sqsubseteq TUnTV, \cdot \exists \neg \exists, \text{est-73}} \boxed{\begin{array}{c} (\exists \exists) n(\epsilon \setminus (\epsilon \setminus R^\circ)) \\ \exists \text{est}(R) \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} \exists (\exists n(\epsilon \setminus (\epsilon \setminus (\epsilon \setminus R^\circ)))) \\ \exists (\exists n(\epsilon \setminus R^\circ)) \end{array}} \quad \text{count of } T \cdot \neg T \setminus \\ \\ \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\begin{array}{c} \text{dom}(\text{est}(R)) \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \\ \\ \xleftarrow{\text{dom}(\text{est}(R)) \sqsubseteq \text{est}(R) \text{ est}(R)^\circ} \boxed{\begin{array}{c} \text{est}(R) \text{ est}(R)^\circ \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{est}(R)^\circ \sqsubseteq \epsilon, \epsilon \epsilon \text{ union} \sqsubseteq \epsilon} \boxed{\begin{array}{c} \text{est}(R) \epsilon \text{est}(R) \\ \text{est}(R) R^\circ \end{array}} \quad \text{UP of est} \end{array}$$

### §11.3. Shortest paths on a cylinder, on lists

(11.3a)

| definition                                                                                            | type                                               | note                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| $F(A, X) = A + A \times X$                                                                            | $\mathcal{A} \rightarrow \mathcal{A}$              | The base functor: one square of the new column, alone or in front of the path so far.                                                  |
| $L = \text{list}^+$                                                                                   | $\mathcal{A} \rightarrow \mathcal{A}$              | A path is a non-empty list of squares, one per column crossed.                                                                         |
| $\alpha$ the initial algebra                                                                          | $F(\text{dCL}(L, E)) \rightarrow \text{dCL}(L, E)$ | $\alpha(5) = [5]$ and $\alpha(1, [5]) = [1, 5]$ : a path is started by one square, or extended by one.                                 |
| $N$<br>the n-tuple relator                                                                            | $\mathcal{A} \rightarrow \mathcal{A}$              | One component per row of a column, so the fold carries n answers at once.                                                              |
| $R \triangleq \text{sum} \leq \text{sum}^\circ$                                                       | $[\text{Nat}] \rightarrow [\text{Nat}]$            | The cost of a path, which the cheapest minimises.                                                                                      |
| setify                                                                                                | $A[n] \rightarrow EA$                              | $\text{setify}(1, 2, 3, 4) = \{1, 2, 3, 4\}$ — which row a component came from is forgotten.                                           |
| moves                                                                                                 | $A[n] \rightarrow E(A[n])$                         | $\text{moves}(x) = \{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.                                         |
| trans                                                                                                 | $E(A[n]) \rightarrow (E A)[n]$                     | $\text{trans}\{(a, b, c), (x, y, z)\} = (\{a, x\}, \{b, y\}, \{c, z\})$ — component k of the result is the set of the k-th components. |
| zip                                                                                                   | $F(a[n]) \rightarrow (F a)[n]$                     | $\text{zip}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$ .          |
| $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$                                                  | $F(EA) \rightarrow EFA$                            | $\text{cp}(1, \{[5], [6], [8]\}) = \{(1, [5]), (1, [6]), (1, [8])\}$ .                                                                 |
| $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{zip } N(\text{cp } E(\alpha))$ | $F(NA, N(E(TA))) \rightarrow N(E(TA))$             | $\text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in (11.3.1b).                                           |
| $\text{paths} \triangleq (\text{gen}) \text{setify union}$                                            | $T(NA) \rightarrow E(TA)$                          | $\text{paths}[(1, 2, 3, 4), (5, 6, 7, 8)]$ is the union of (11.3.1b)'s four sets: 12 paths, 3 from each entry row.                     |
| the specification<br>$\text{paths est}(R)$                                                            | $[\text{Nat}[n]] \rightarrow [\text{Nat}]$         | A cheapest path from the entry side to the exit side.                                                                                  |

§11.3.1.  $N(E(L A))$ 

$$\begin{array}{l} 1 \quad 5 \\ 2 \quad 6 \\ 3 \quad 7 \\ 4 \quad 8 \end{array} \quad [(1, 2, 3, 4), (5, 6, 7, 8)] : L \ N \ \text{Nat}$$

(11.3.1a)

$(\text{gen})[(5, 6, 7, 8)] = (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$

(11.3.1b)

$$\begin{aligned} \text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = \\ ( \{ [1, 5], [1, 6], [1, 8] \}, \\ \{ [2, 5], [2, 6], [2, 7] \}, \\ \{ [3, 6], [3, 7], [3, 8] \}, \\ \{ [4, 5], [4, 7], [4, 8] \} ) \end{aligned}$$

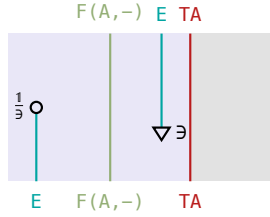
(11.3.1c)

|   |                                                        |
|---|--------------------------------------------------------|
| N | the 4-tuple, one component per row of the entry column |
| E | the set of paths that can start in that row            |
| L | a path is the non-empty list of the squares it crosses |

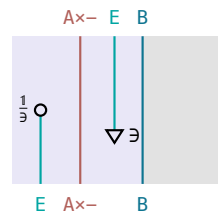
row 1's set has no  $[1,7]$  — from row 1 only rows 4, 1 and 2 are reachable — and it does have  $[1,8]$ , because the cylinder glues the bottom row to the top. That is **moves trans N(union)** of  $\triangleq$  **paths**: component  $k$  collects the paths of rows  $k-1, k, k+1$ .

### §11.3.2. The cross product $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$

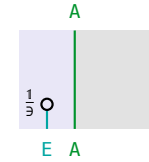
(11.3.2a)



cp on all of  $F(A, E(L A))$



the  $A \times -$  summand:  $\exists$  picks one path,  $1\% \exists$  collects the results



the  $A$  summand, no  $E$  to distribute:  $1\% \exists$  alone,  $a \mapsto \{a\}$

(11.3.2b)

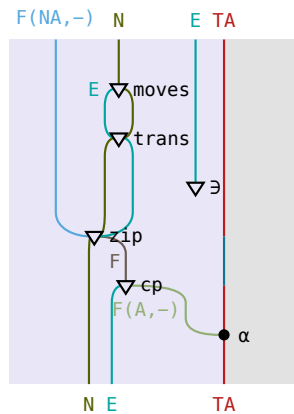
| type                                                              | note                                                                                                   |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $F(A, -) = A + A \times -$                                        | one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$   |
| $\exists : E(L A) \rightarrow L A$                                | the $\exists$ inside $F(1, \exists)$ : a set of paths, one of them — $L A$ is one path, $A$ one square |
| $F(1, \exists) = 1 + 1 \times \exists$                            | $: A + A \times E(L A) \rightarrow A + A \times L A$ , the relator acting on each summand              |
| $\exists : E(A + A \times L A) \rightarrow A + A \times L A$      | the $\exists$ under the bar: a set of cells, one of them — a different $\exists$                       |
| $\text{cp} : A + A \times E(L A) \rightarrow E(A + A \times L A)$ | $\frac{R}{\exists}$ turns $R: X \rightarrow Y$ into $X \rightarrow E Y$                                |
| $\text{cp} = \frac{1}{\exists} E(F(1, \exists))$                  | what the picture draws: the unit makes the outer $E$ , leaving one $\exists$ — pow-laws                |

(11.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$  the right summand  
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$   $1$  keeps the square  $1$ ,  $\exists$  picks one path  
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$  one output per path in  $w$   
 $\text{cp}(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$  the three of them, collected

### §11.3.3. $\text{gen} = F(1, \text{moves trans N(union)}) \text{ zip N(cp P}(\alpha))$

(11.3.3a)



(11.3.3b)

```

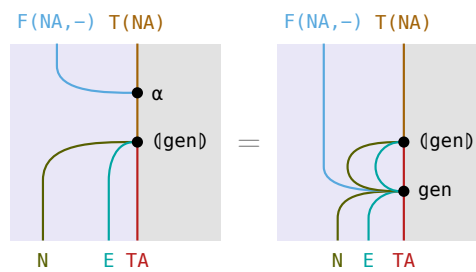
u = ((1,2,3,4),({[5]},{[6]},{[7]},{[8]})) : F(N A,N(E(L A)))
F(1,moves trans N(union))                  1 keeps the column, the path SETS move
moves({[5]},{[6]},{[7]},{[8]})
= ({[6]},{[7]},{[8]},{[5]}),
  ({[5]},{[6]},{[7]},{[8]}),
  ({[8]},{[5]},{[6]},{[7]})} : E(N(E(L A))) down, unmoved, up
trans(that)
= ({[6]},{[5]},{[8]},{[7]},{[6]},{[5]}},
  {[8]},{[7]},{[6]},{[5]},{[8]},{[7]})} : N(E(E(L A))) row k gets rows k-1, k, k+1
N(union)(that)
= ({[5],[6],[8]},{[5],[6],[7]},
  {[6],[7],[8]},{[5],[7],[8]}) : N(E(L A)) every path into row k, none dropped
zip(that)
= ((1,{[5],[6],[8]}),(2,{[5],[6],[7]}),
  (3,{[6],[7],[8]}),(4,{[5],[7],[8]})) : N(F(A,E(L A))) each row: its square, and the
                                                                paths it may be put in front of

N(cp P(α))(that)
= ({[1,5],[1,6],[1,8]},{[2,5],[2,6],[2,7]},
  {[3,6],[3,7],[3,8]},{[4,5],[4,7],[4,8]}) : N(E(L A)) cp pairs the square with each path,
                                                                α prefixes it: α(1,[5])=[1,5]

```

#### §11.3.4. gen is an F-algebra; (gen): α(gen)=F(1,(gen))gen

(11.3.4a)



α puts the column back on the list, then the fold reads all of it

the fold reads the rest under F, then one gen puts the column in front

$$xs = [(1,2,3,4),(5,6,7,8)] = \alpha((1,2,3,4),[(5,6,7,8)]) : L(N A)$$

(11.3.4b)

$$(\text{gen})(xs) = \text{gen}((1,2,3,4), (\text{gen})[(5,6,7,8)])$$

both halves are in (11.3.1b): the tail folds to  $(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$ , one path per row and each of them one square long, and **gen** on it is (11.3.3b)'s walk.

#### §11.3.5. Q : F(N A,N(L A))→N(L A), an F(N A,-)-algebra: Q=F(1,moves trans N(est(R))) zip N(α)

(11.3.5a)

```

Q : F(N A,N(L A)) → N(L A)                  the fold's algebra: a new column, one path per row

u = ((1,2,3,4),([5],[6],[7],[8])) : F(N A,N(L A))
F(1,moves trans N(est(R)))              1 keeps the column, the paths move
moves([5],[6],[7],[8])
= ({[6],[7],[8],[5]},
  {[5],[6],[7],[8]},
  {[8],[5],[6],[7]}) : E(N(L A)) down, unmoved, up
trans(that)
= ({[6],[5],[8]},{[7],[6],[5]},
  {[8],[7],[6]},{[5],[8],[7]}) : N(E(L A)) row k gets rows k-1, k, k+1
N(est(R))(that)
= ([5],[5],[6],[5]) : N(L A) the cheapest into each row –
                                                                chosen BEFORE the new square

zip(that)
= ((1,[5]),(2,[5]),(3,[6]),(4,[5])) : N(F(A,L A)) each row: its square, and the one
                                                                predecessor that survived

N(α)(that)
= ([1,5],[2,5],[3,6],[4,5]) : N(L A) α(1,[5])=[1,5]

```

| $Q = [N(\text{wrap}), (1 \times \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\text{cons})]$ (11.3.5b)                                                                                                                                                                                                                                                                                  |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| at the last column $Q$ starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra (11.3.5c)'s last step folds                                                                                                                                                                                          |               |
| circuit — the fork is $F(NA, N(LA)) = NA + NA \times N(LA)$                                                                                                                                                                                                                                                                                                                                    | Hinze–Marsden |
| $F(N A) (N (T A)) \text{---} \boxed{Q} \text{---} N(T A)$                                                                                                                                                                                                                                                                                                                                      |               |
| $= FN(T A) \text{---} \boxed{F(1, \text{moves trans } N(\text{est}(R)))} \text{---} FN(T A) \text{---} \boxed{\text{zip}} \text{---} N(F A (T A)) \text{---} \boxed{N(\alpha)} \text{---} N(T A)$ <p>the fusion condition (11.3.5d) read as a definition,</p>                                                                                                                                  |               |
| $= F(N A) (N (T A)) \text{---} \left[ \begin{array}{l} N(p_{\circ}) \text{---} \boxed{N(\text{wrapA})} \text{---} N(T A) \\ q \text{---} \boxed{\text{pm}} \text{---} q \text{---} \boxed{\text{zip}'} \text{---} N(q_{\circ}) \text{---} \boxed{N(\text{consA})} \text{---} N(T A) \end{array} \right]$ <p><math>\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]</math></p> |               |

[illegible]

| $\text{gen } N(\text{est}(R)) \exists F(1, N(\text{est}(R))) Q$ <span style="float: right;">(11.3.5d)</span>                                                                                                                                                                                                                                                       |                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| fusion: the condition for fusion in (11.3.5c)'s last step, used to derive a definition of $Q$ .                                                                                                                                                                                                                                                                    |                                                                                                                                    |
| <b>circuit</b> <div style="text-align: center; margin-top: 20px;"> <math display="block">F(N\ A)\ (N\ (E\ (T\ A))) \text{---} \boxed{\text{gen}} \text{---} N(E\ (T\ A)) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A)</math> </div>                                                                                                                      | <b>Hinze–Marsden</b> <div style="text-align: center; margin-top: 10px;"> <math display="block">F(N\ A, N(E\ (T\ A)))</math> </div> |
| <div style="display: flex; align-items: center; margin-top: 20px;"> <div style="margin-right: 10px;"> <math>\sqsubseteq</math> </div> <div> <math display="block">F(N\ E\ (T\ A)) \text{---} \boxed{F(1, N(\text{est}(R)))} \text{---} F(N\ (T\ A)) \text{---} \boxed{Q} \text{---} N(T\ A)</math> </div> </div> <p>(7.13), then zip, trans, moves lax natural</p> | <div style="text-align: center; margin-top: 10px;"> <math display="block">F(NA, -) \quad N \quad E \quad TA</math> </div>          |

(7.13) is  $F(1, \text{est}(R)) \alpha \sqsubseteq_{\text{cp}} P(\alpha) \text{ est}(R)$ , mon-thm71 at the map  $\alpha$  with  $\frac{F(1, \emptyset) \alpha}{\exists} =_{\text{cp}} P(\alpha)$ : extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.